Online Forest Disturbance Detection at the Sub-Annual Scale Using Spatial Context From Sparse Landsat Time Series

Ling Wu, Xiangnan Liu, Meiling Liu, Jinghui Yang, Lihong Zhu, and Botian Zhou

***Abstract*— Mapping forest disturbances using dense time series can timely identify disturbances at the subannual scale. However, these change detection methods using dense time series may be infeasible when not enough temporal observations are available. In this article, an online change detection algorithm that identifies forest disturbances at a subannual scale using spatial context from the sparse Landsat time series was proposed. First, the spatial normalized index that removed forest seasonality was prepared for establishing a simplified model instead of the harmonic model, thereby reducing the requirements for a high temporal frequency of clear observations for model initialization. Second, by using the spatial errors model to establish the simplified model, the normally distributed residual time series that removed the spatial autocorrelation were obtained. Third, the spatial statistic *t* time series transformed from residual time series within a 3 × 3 spatial window were subsequently subjected to the exponentially weighted moving average *t* chart (EWMA-t), which is a statistical process control chart for a short cycle corresponding to sparse Landsat time series. Fourth, disturbed pixels were labeled if the chart values persistently deviated from the control limits of the chart. The proposed algorithm was applied to a subtropical forest with low Landsat data availability and yielded an overall accuracy of 86% in the spatial domain and temporal accuracy of 93.7%, achieving accurate and timely identification of forest disturbances. The proposed method called the EWMA-t change detection (EWMATCD) algorithm provides an alternative for disturbance detection at the subannual scale in regions with low data availability.**

***Index Terms*— Disturbance detection, exponentially weighted moving average *t* chart (EWMA-t chart), sparse Landsat time series, spatial context.**

I. Introduction

THE monitoring of forest disturbances has received much research attention due to the increasing human activities in recent decades. As important repetitive observation data of the Earth's surface, satellite images provide an ideal source of data for monitoring forest disturbance at the global scale [1], [2]. Compared with monitoring forest disturbances by comparing the differences between two images at different time steps, mapping forest disturbance using change detection algorithms for time series has received increasing applications in recent years [3]–[15] because such dynamic and iterative process not only improves operational efficiency but also guarantees mapping accuracy by fully utilizing the temporal context.

When detecting disturbances using time series, the seasonal variations among satellite images across various time steps, which are induced by phenological changes and sun angle differences, greatly interfere in disturbance detection and should be eliminated. Kennedy *et al.* [4] utilized one image per year in the same season to minimize the phenological changes and sun angle differences in the LandTrendr algorithm. Nevertheless, the change detection accuracy depends on the availability of ideal images in the same season because of cloud cover and shadows, which leads to temporal delays in disturbance detection at the subannual scale. Dense time series (much more than one image per year) allow users to describe seasonal variations, which is useful for achieving disturbance detection at the subannual scale. Some methods detect changes at the subannual scale with dense time series by considering seasonal variations using a harmonic model [6], [7], [16]–[19]. The seasonal and trend components that are decomposed from dense MODIS time series using the Breaks For Additive Seasonal and Trend (BFAST) algorithm are used to capture the seasonal variations and trend, respectively, and the detected breaks represent the occurrence of disturbances [16], [17]. However, the MODIS time series with a coarse spatial resolution have limitations in detecting small changes [20]. Therefore, the use of time series with a medium spatial resolution, such as Landsat, in change detection has received increasing attention. The continuous change detection and classification (CCDC) algorithm proposed by Zhu and Woodcock [18] adopts a single-harmonic model to describe the seasonal trajectories of different land cover categories, detects the land surface change, and classifies land cover categories after change detection by using all available Landsat images. Given the infeasibility of capturing complex subannual changes (e.g., farmlands with double- or triple-crop patterns within a year) using the single-harmonic model in CCDC, Chen *et al.* [19] used a multiharmonic model instead of a single-harmonic model for cropland

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change detection. By using the exponentially weighted moving average (EWMA) chart, Brooks *et al.* [6] presented the EWMA change detection (EWMACD) algorithm that utilizes the residuals from harmonic regression over years of Landsat time series to identify both deforestation and forest subtle disturbances. Most deseasoning change detection methods that utilize dense time series, especially online change detection methods, heavily focus on information in the temporal domain, which means that the change detection accuracy is excessively dependent on the quantity and quality of time-series data in this domain. For example, the minimum training period in the BFAST monitor algorithm should not be shorter than two years when MODIS time series with a 16-day temporal resolution or Landsat time series that are irregularly spaced in the temporal domain are used [10], [17]. Moreover, no disturbances should occur during the training period for online change detection. To reduce the likelihood for these disturbances to occur in the CCDC and dynamic EWMACD (Edyn) algorithms [7], [18], a minimum training length that is at least three times larger than the number ($\geq$3) of fitting parameters in the harmonic model was determined, and then, from the beginning of the training period, the harmonic model was successively fitted within a moving window with the minimum training length until the accuracy requirement or the maximum training length (twice as long as the length of the minimum training period) is met. However, when the available observations are insufficient, the time span of the training period is extended to accurately fit the harmonic model, thereby increasing the likelihood for disturbances to occur during the training period. For example, the number of clear observations for model initialization may be insufficient within the large time interval (e.g., longer than two years) between two disturbances. This situation is commonly observed in rapid development regions with frequent disturbances or those with frequent cloud cover and shadows, and poses additional difficulties in detecting multiple disturbances online [21]. Therefore, the high requirements for the high temporal frequency of clear observations and the lack of any disturbances during the training period with a large time span render those methods that use the harmonic model inapplicable in regions with frequent cloud cover and shadows.

The acquisition of Landsat images at an irregular interval may also lead to missing images in "key dates" of a year required by harmonic regression [22], and a seasonal model over-fitting may occur when not enough temporal observations are available due to frequent cloud cover and shadows [7], [12], [18]. Therefore, the subannual seasonal variations may not be well described because of the irregular sampling interval and the limited number of clear observations, thereby leading to the infeasibility of detecting disturbances on the premise of minimizing the influence of seasonal variations. One solution to the irregular sampling interval and the limited number of clear observations is to densify the Landsat time series and then describe the seasonal variations. Platt *et al.* [23] interpolated the missing images to create a monthly Landsat time series for land cover detection, and Wu *et al.* [24] monitored the spatiotemporal dynamics of forests using monthly synthetic Landsat time series that are reconstructed by integrating two spatiotemporal interpolation algorithms into a spatiotemporal data fusion algorithm. The synthetic Landsat images effectively fill the excessive temporal gaps and can be used to describe the seasonal variations for change detection. However, change detection using synthetic dense Landsat time series still relies on the quantity and quality of actual Landsat images, which determines the reconstruction accuracy of time series [25]. In addition, most of the currently available interpolation or data fusion algorithms reconstruct the contaminated pixels without considering potential disturbances (e.g., land cover change) and are, therefore, incorrectly regarded as stable, which may lead to delays in disturbance detection [24].

When the clear observations in the temporal domain are not enough, an alternative is to lower the temporal frequency requirement of clear observations for model initialization on the premise of minimizing the influences of seasonal variations. This approach can reduce both the length of the training period and the likelihood for disturbances to occur during the training period. Given that fitting the harmonic model for describing seasonal variations plays a key role in satisfying the requirement for a high quantity of clear observations, one effective strategy for minimizing the influences of seasonal variations without satisfying such requirement is to utilize normalized indices that integrate the spatial context, such as the disturbance index (DI) [5], [26], [27], the integrated forest Z-score (IFZ) [3], [28], or the normalized NDVI [12], [29]. DI and IFZ minimize the seasonal variations by calculating the Z-score using the global mean and standard deviation of forest pixel values across Tasseled Cap bands or spectral bands. Under the assumption that the seasonal variations of the focal pixel are similar to those of the spatially neighboring pixels, Hamunyela *et al.* [12] used the spatially normalized NDVI (SNDVI), which divides the value of the focal pixel by the median value of the pixels that are above or equal to the 90th percentile within an optimal spatial moving window to reduce the seasonal variations. Afterward, the clear observations at only six time steps can initialize the simple model that only includes the constant term instead of the harmonic model, thereby reducing the length of the training period and the likelihood for disturbances to occur within a short period.

Online change detection methods, such as the BFAST monitor and EWMACD algorithms, detect disturbances by establishing a boundary of the observations' empirical fluctuation or control limits (CLs) and then judging whether the incoming observations cross such boundary or limits. These boundaries or CLs depend on the estimated standard deviation of residuals between the model fitting values and the observed values during the training period. When the aforementioned strategy of initializing a simple model using a limited number of temporal observations by integrating the spatial context is adopted, the standard deviation estimator, which is based on a limited number of temporal observations instead of taking into account the high temporal frequency of clear observations during the training period, may be inaccurate due to sparse observations. Therefore, the performance of the existing online change detection methods deteriorates. In this case, establishing a reliable boundary of empirical fluctuation or CLs based on a limited number of temporal observations

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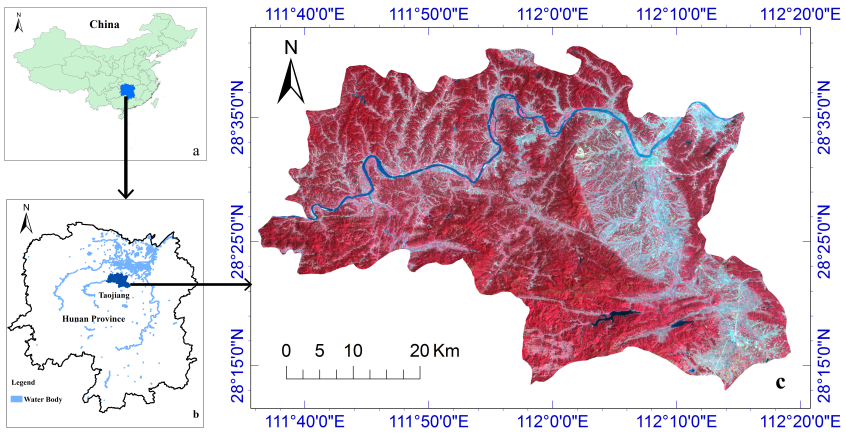


Fig. 1. Geographic location and Landsat image of the study area: (a) general location within China, (b) within Hunan province, and (c) Landsat 5 TM image (RGB = near-infrared, red, and green band).

is critical when applying online change detection methods in regions with frequent cloud cover and shadows. In statistical process control, some control chart techniques, such as Q or t charts [30]–[33], are proposed for a short production run and have been proven robust against errors in estimating the process standard deviation. Take the combination of EWMA chart and t chart (i.e., EWMA t chart (EWMA-t) control chart) for example. Instead of the single sample at a time used in the EWMA control chart, the statistic t computed using multiple samples at the same time (a subgroup) is used to establish reliable CLs [31]. This technique does not require a large number of subgroups, thereby suiting the short production run and quickly detecting out-of-control signals [33]. In sum, the EWMA-t control chart may be feasible for detecting disturbances using spatiotemporal observations at a small number of time steps and is, thereby, suitable for those regions with a low temporal frequency of clear observations.

In this article, we examine how online detection of forest disturbances can be detected at the subannual scale using sparse Landsat time series. In this article, the sparse Landsat time series are defined such that the number (>1) of clear temporal observations per year is too small to establish a reliable harmonic model for describing seasonal variations. Sparse Landsat time series with a low temporal frequency of clear observations are common in perpetually cloudy tropical or subtropical forests [34]. To achieve the aforementioned objective, we introduced the EWMA-t change detection (EWMATCD) algorithm, which detects forest disturbance at the subannual scale with a low temporal frequency of clear Landsat observations by integrating the spatial context in three steps. First, normalized NDVI time series are calculated by integrating spatial neighborhood information to minimize the influence of seasonal variations, simplify the model form, and reduce the temporal frequency requirement of clear observations for model initialization. Second, the regression residual time series with an independent normal distribution in the temporal–spatial domain are computed using the spatial errors model (SEM) to establish the simplified fitting model. Third, the spatial statistic t time series transformed from the residual time series within a 3×3 spatial window are subjected to the EWMA-t chart for disturbance detection. Disturbances are labeled when the chart values persistently deviate from the CLs of the EWMA-t chart.

II. Study Area and Data

A. Study Area

The study area, Taojiang county, is located in the mid-northern part of Hunan Province and is the key forest county of China (see Fig. 1). Mountains, hills, and hillocks account for over 70% of Taojiang, and the county reported a 57% forest cover in 2006. As the main disturbance in Taojiang, deforestation for getting useful timber was treated as the disturbance detection target in this study. Taojiang county is located in a subtropical zone belonging to a continental monsoon humid climate region. Therefore, the rainy climate in this area leads to the contamination of satellite data by clouds, cloud shadows, and atmospheric scatter, hence providing an ideal context for testing the feasibility of online forest disturbance detection at the subannual scale using sparse Landsat time series.

B. Landsat Data and Forest Benchmark Mask

NDVI time series from 2001 to 2018 were collected from Landsat TM5, ETM+7, and OLI8 surface reflectance (SR) images obtained from the U.S. Geological Service (USGS). A total of 102 Landsat images with less than 20% pixels contaminated by clouds, cloud shadows, and SLC-off gaps were collected in the study area, in which the average number of clear observations per year is close to 6 (see Fig. 2). A spatiotemporal interpolation method called NSPI [35] was used to fill in the SLC-off gaps in the ETM+7 images since 2003. Clouds and cloud shadows were masked out according to the function of mask products (Fmask), which were distributed together with Landsat SR products. A forest benchmark mask was also generated for masking the nonforest area based on the Landsat classification image taken on April 14, 2003, (the maximum training date for the most pixels), thereby ensuring that the analysis targets are forest pixels at the beginning of the monitoring period and minimizing the occurrence probability of forest disturbances during the training period.

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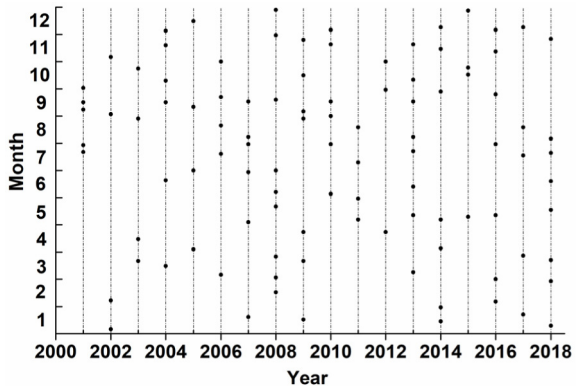


Fig. 2. Temporal distribution of available images with less than 20% of contaminated pixels between 2001 and 2018.

C. Validation Data

Because of lacking other remote sensing images with both higher temporal and finer spatial resolution than Landsat images during a long time span from 2001 to 2018, the primary and optimal sources for assessing forest disturbance detection accuracy based on Landsat time series are the Landsat images themselves [18], [36]. The high spatial resolution images acquired from Google Earth were treated as supplements to help manually interpret the disturbances. The random stratified sample design proposed by Zhu and Woodcock [18] was used to select 500 reference patches with 3×3 pixels as validation data from the result of disturbance detection using the proposed EWMATCD method. A total of 250 reference patches were used to represent those areas with undisturbed forests, and another 250 reference patches were used to represent the disturbed forests. For each validation data, we visually examined Landsat images and high spatial resolution images to determine whether and when forest disturbances occurred in a certain location.

III. Methodology

Fig. 3 illustrates the general workflow of online forest disturbance detection adopted in this work. First, the classification map was obtained using the random forest algorithm, and the nonforest areas were masked out. Second, the SNDVI for reducing seasonal variations was calculated using the spatial context to reduce the high temporal frequency requirement of clear observations for model initialization (see Section III-B). Third, for satisfying the requirement that the data used in the change detection method (EWMA-t chart) should follow an independent normal distribution in both the temporal and spatial domains, the SEM model was fitted using temporal–spatial observations with 3×3 spatial segments multiplied by the temporal length of the training period, and normally distributed residual time series were obtained by subtracting the observations from the predicted values obtained from the SEM model (see Section III-C). Fourth, given that only a limited number of images were used to establish the boundary of empirical fluctuation in the training period, statistic T_i, which can improve the reliability of the established boundary, was computed based on the observations reported within the spatial window at time step i. Afterward, the T_i time series were inputted into the EWMA-t chart, which was developed for statistical process control in the short cycle (corresponding to the low temporal frequency of clear observations in this study), to compute the EWMA-t values. Disturbances are detected when the EWMA-t values persistently crossed the CLs (see Section III-D). The critical procedures are discussed in detail in the following.

A. EWMACD Algorithm

We first describe EWMACD [6] before explaining the proposed EWMATCD. Given a time series of clear observations with length N, $t = \{t_1, t_2, \ldots, t_N\}$, the following harmonic model was established in the training period:

$$y_i = a_0 + \sum_{j=1}^{k}\left(b_j \cos\left(\frac{2\pi j}{D}t_i\right) + c_j \sin\left(\frac{2\pi j}{D}t_i\right)\right) + \varepsilon_i \quad (1)$$

where a_0 is the intercept coefficient, k is the order of harmonic function, b_j and c_j are the coefficients of cosine and sine harmonics, respectively, ε_i is the regression residual at Julian date t_i, $i = \{1, 2, \ldots, N\}$, and D is the number of days per year. These model coefficients are estimated using the ordinary least-squares (OLS) method.

The EWMA values for the entire time series were calculated as

$$\mathrm{EWMA}_i = (1-\lambda)\mathrm{EWMA}_{i-1} + \lambda\varepsilon_i \quad (2)$$

where the smoothing parameter λ, with values ranging from 0 to 1, is used to measure the retrospectives of the chart. A λ value close to 1 indicates that little weights are assigned to historical data, whereas a λ value close to 0 indicates that the result of EWMA_i mainly relies on previous values EWMA_{i-1}. EWMA_1 was set to 0.

Suppose that the regression residuals ε_i have a stable mean of 0, and the CLs CL_i for signaling disturbances were computed as

$$\mathrm{CL}_i = 0 \pm Ls\sqrt{\frac{\lambda}{2-\lambda}[1-(1-\lambda)^{2i}]} \quad (3)$$

where L is the CL constant and s is the standard deviation of all residuals ε_i in the training period.

Then, the index EWMACD was calculated by dividing the EWMA values by the CL values for signaling disturbances

$$\mathrm{EWMACD}_i = \mathrm{sign}(\mathrm{EWMA}_i) \cdot \mathrm{int}\left[\left|\frac{\mathrm{EWMA}_i}{\mathrm{CL}_i}\right|\right] \quad (4)$$

where sign() is used to acquire the plus-minus sign of EWMA_i, which describes the direction of deviation from the empirical fluctuation, || represents the absolute value of EWMA_i/CL, and int[] represents the integer of this absolute value. When EWMA_i exceeds the lower CLs CL_i, the EWMACD value is less than 0, and the EWMA chart signals a disturbance; otherwise, the process remains in control and no disturbance occurred when the EWMACD value equals 0.

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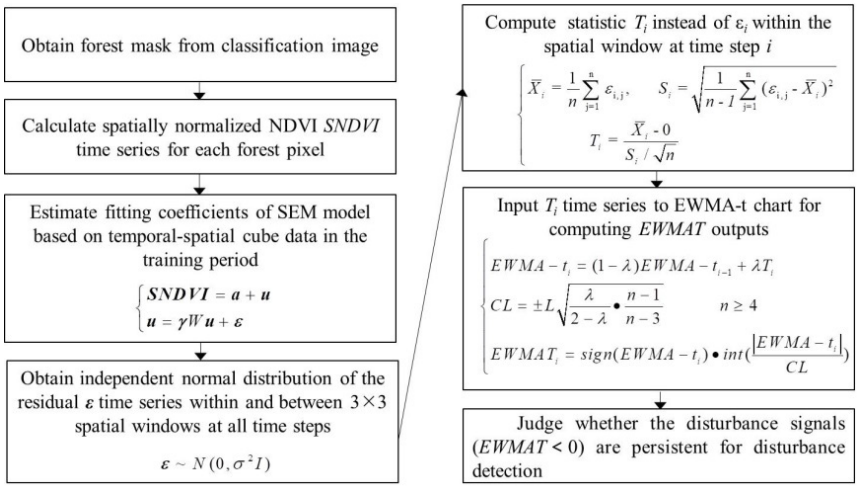


Fig. 3. Flowchart of the EWMATCD algorithm.

B. Calculation of Spatially Normalized NDVI for Model Initialization in Sparse Landsat Time Series

Building upon the spatially normalized approach proposed in [12], the SNDVI for reducing phenological variations was calculated using the spatial context as follows. First, the median value $\mathrm{NDVI}_{\mathrm{median}}$ was calculated from the values of those pixels that exceed a certain percentile P within a spatial window, the focal pixel of which is the target pixel being normalized. Second, the value of the focal pixel ($\mathrm{NDVI}_{\mathrm{pixel}}$) was divided by $\mathrm{NDVI}_{\mathrm{median}}$. This spatially normalized approach is based on the assumption that the upper tail of pixels within the spatial window represents forest pixels, whose temporal dynamics are similar to those of the focal pixel. Following this assumption, the ratio of $\mathrm{NDVI}_{\mathrm{pixel}}$ to $\mathrm{NDVI}_{\mathrm{median}}$ can remove the seasonal variations. Therefore, SNDVI time series were obtained after repeating the normalization procedure for each forest pixel within the classification image and each time step from 2001 to 2018 (including the model training and monitoring periods). The spatial window size (SWS) was defined as

$$\mathrm{SWS} = \mathrm{int}\left[\left(\sqrt{M/(1-P)}+1\right)/2\right] \times 2 + 1 \tag{5}$$

where M is the number of the upper tail of pixels within the spatial window. To guarantee the statistical significance of the statistical index $\mathrm{NDVI}_{\mathrm{median}}$, the value of M should not be less than 30. Therefore, M was set to 30 in this study, whereas P was set to 90%. Thus, the calculated SWS was 19. Other values of M and P can also be used to calculate $\mathrm{NDVI}_{\mathrm{median}}$.

Given that SNDVI time series do not include the seasonal variations described by the harmonic model, a simple linear model, instead of the harmonic model in (1) established during the model training period, was constructed as

$$\mathrm{SNDVI}_i = a + u_i \tag{6}$$

where a is estimated using OLS and u_i represents the regression residual at Julian date t_i. A minimum of six clear observations can fit the simple linear model [12], thereby reducing the length of the training period for those regions with a low temporal frequency of clear observations.

C. Acquisition of Independently Normally Distributed Residual Time Series Using SEM Model

In this article, forest disturbance was detected using the EWMA-t chart, in which the observations in both the temporal and spatial domains are assumed to follow an independently normal distribution. The input data of the EWMA-t chart consist of n spatially neighboring time series within a 3×3 spatial window. The residuals within each single time series follow an independently normal distribution with $\mu 0 = 0$ when these residuals are calculated using the simple linear model (6). However, the assumption that the n spatially adjacent residuals of different time series follow an independently normal distribution at the same time step may not be supported because strong spatial correlations are very likely to be observed among these residuals [37]. Therefore, before applying the EWMA-t chart to the regression residuals for disturbance detection, the spatial correlation should be removed using SEM (7) instead of (6)

$$\begin{aligned} \boldsymbol{SNDVI} &= \boldsymbol{a} + \boldsymbol{u} \\ \boldsymbol{u} &= \gamma W \boldsymbol{u} + \boldsymbol{\varepsilon} \\ \boldsymbol{\varepsilon} &\sim N(0, \sigma^2 I) \end{aligned} \tag{7}$$

where $\boldsymbol{SNDVI}$ represents the matrix of SNDVI observations with size N (length of time series) rows multiplied by n (number of spatially neighboring forest pixels) columns, $\boldsymbol{a}$ denotes n regression coefficients a of n neighboring forest time series in (6), $\boldsymbol{u}$ is the matrix of spatial correlated regression residuals u_i in (6), γ is a spatial dependence parameter that can be estimated via maximum likelihood, W is a spatial weight matrix filled with 1 for neighboring forest pixels and 0 otherwise in the 3×3 spatial window, ε is the matrix of independently normally distributed residuals with the size $N \times n$, and I is the identity matrix. The SEM model can be rewritten as

$$\boldsymbol{SNDVI} = \boldsymbol{a} + (\boldsymbol{I} - \gamma W)^{-1}\boldsymbol{\varepsilon}. \tag{8}$$

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Therefore, the regression residuals ε between n time series are spatially independent, and the EWMA-t chart can be applied to ε instead of the regression residuals u_i of the simple linear model (6) to detect disturbances.

The minimum length of the training period (six time steps were adopted in this study) reduces the likelihood for a disturbance to occur during the training period. To avoid the extreme case that a disturbance occurs during the first six time steps, we followed the approach of Brooks *et al.* [7] and Zhu and Woodcock [18] and used a moving window with a length of 6 to remove the observations at the first time step within the window and then added the new observations at the next time step after the last time step within the window to successively initialize model until the fitting SEM model meets the minimum fit accuracy requirement $R^2_{\min}$ or the moving window reaches the maximum training date. $R^2_{\min}$ was set to 0.7, and the maximum training date was defined as the time step in the time series that is twice longer than the endpoint of the first training period. Given that some observations in the training period may be outliers or contaminated pixels that were not captured by the previous Fmask algorithm, in each model fitting process within the moving window, we followed two fitting processes. Specifically, the model was initially fitted using all observations at six time steps, and those observations whose fitting residuals ε cross the standard deviation estimator of all residuals values three times were excluded. Then, the remaining clear observations were used to fit the model again. After its establishment in the training period, the SEM model was used as the basis for disturbance detection in the monitoring period.

D. Disturbance Detection at the Subannual Scale Considering the Spatial Context Using the EWMA-t Chart

In the EWMA-t control chart, we consider the matrix of the regression residuals ε calculated by (8) $\{\varepsilon_{1,n}, \varepsilon_{2,n}, \ldots, \varepsilon_{i,n}\}$, at Julian date t_i, where n is the number of forest pixels within the 3×3 spatial window. The residuals ε follow an independently normal distribution with mean $\mu 0 = 0$ and standard deviation $\sigma 0$ within and between subgroups $\varepsilon_{i,n}$.

The subgroup mean $\bar{X}_i$ and standard deviation S_i at Julian date t_i were defined as

$$\bar{X}_i = \frac{1}{n}\sum_{j=1}^{n} \varepsilon_{i,j}, \quad S_i = \sqrt{\frac{1}{n-1}\sum_{j=1}^{n}(\varepsilon_{i,j} - \bar{X}_i)^2}. \tag{9}$$

To improve robustness against errors in estimating the process standard deviation in the EWMA chart, the plotting statistic T_i, instead of the residual ε_i in (2), was defined as

$$T_i = \frac{\bar{X}_i - 0}{S_i/\sqrt{n}}. \tag{10}$$

Therefore, the EWMA-t statistic EWMA-t_i at Julian date t_i was formulated as

$$\text{EWMA-t}_i = (1-\lambda)\text{EWMA-t}_{i-1} + \lambda T_i. \tag{11}$$

To optimize λ and to ensure the efficiency of the EWMA-t chart in the study area, the EWMA-t chart was repeatedly executed on a representative subset of the study area by modifying the value of λ from 0.1 to 1 with a step size of 0.1. On the basis of the principle of effectively detecting disturbance and reducing false positives, the value of λ was set to 0.5. The initial value EWMA-t_0 is usually set to 0. The pixels within the spatial window were marked as disturbed when the chart value EWMA-t_i exceeds the lower CLs in (12) [33]. The CLs were determined by λ, n, and the CL constant L. The value of L was empirically set to 1.5

$$\text{CL} = \pm L\sqrt{\frac{\lambda}{2-\lambda}\cdot\frac{n-1}{n-3}}, \quad n \geq 4. \tag{12}$$

Similar to (4), the EWMAT index was computed as follows to signal disturbances:

$$\text{EWMAT}_i = \text{sign}(\text{EWMA-t}_i)\cdot \text{int}\left[\left|\frac{\text{EWMA-t}_i}{\text{CL}}\right|\right]. \tag{13}$$

A forest is considered undisturbed when its EWMAT value equals 0, and an EWMAT value of less than 0 indicates a possible disturbance. A disturbance is recorded when potential disturbances are identified at multiple consecutive time steps in the same direction, which can effectively reduce the false positives (commission errors) resulting from ephemeral factors, such as undetected clouds, cloud shadows, and snow, in the monitoring period [6], [18]. The quantitative measurement of *persistence* proposed by Brooks *et al.* [7], which indicates the number of multiple consecutive time steps, was used to determine disturbances and set to 0.5 of the mean number of time steps per year in the time series. For example, if the length N of the entire clear time series from 2001 to 2018 for one 3×3 spatial window (patch) is 102, then the required number of consecutive signals for determining disturbance is rounded to 3 according to the *persistence* index (0.5) and the mean number ($102/18 = 5.67$) of time steps per year in this patch.

Instead of each forest pixel, the forest pixels within a patch were treated as a whole to detect disturbance using the EWMA-t chart. Forest pixels of less than 4 within the spatial window counted from the forest benchmark mask were excluded from the analysis and masked out following the sample size n requirements specified in (12). The EWMA-type chart can detect subtle disturbances [6]. Therefore, a subtle disturbance may also be detected even if parts of forest pixels within the spatial window are disturbed. Meanwhile, given the small size of the spatial window, the probability for the rest of the neighboring forests to be disturbed at the subsequent time steps is high if a disturbance occurs in some forest pixels. Therefore, we treated the time step at which forest disturbance first occurred in some forest pixels as the occurrence time of forest disturbance within the spatial window. This proposed algorithm only detects and records the first disturbance for all forest pixels within each spatial window if the forests in the spatial window have experienced multiple disturbances in the entire time series.

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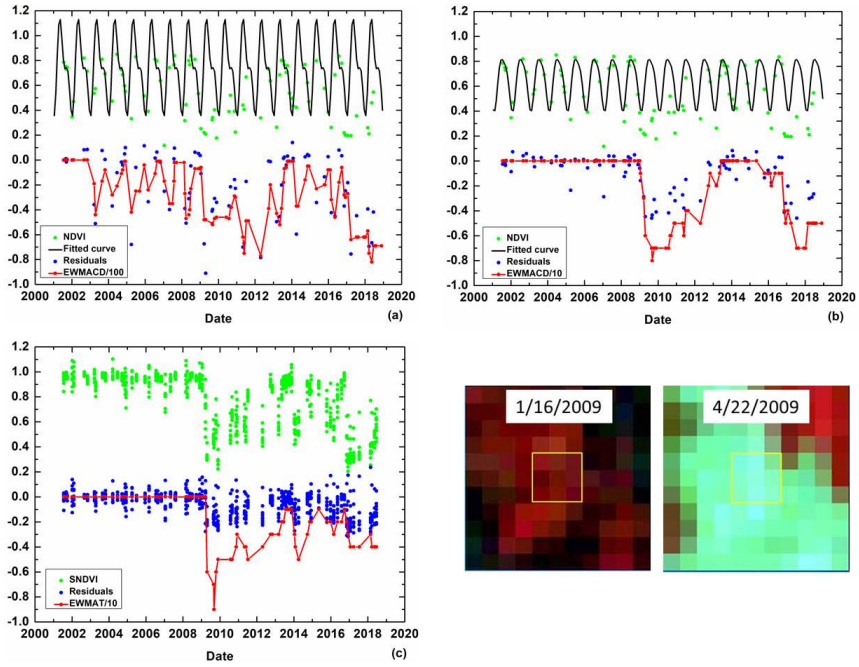


Fig. 4. Disturbance detection for forest pixels within a 3 × 3 spatial window (yellow boxes in Landsat images before and after the disturbance event) using the EWMACD and EWMATCD algorithms: EWMACD algorithm with the number of training data equaling (a) 6 and (b) 15; (c) EWMATCD algorithm with the number of training data equaling 3 × 3 (spatial window) × 6 (time steps).

IV. Results

A. Feasibility Analysis of Disturbance Detection in Sparse Landsat Time Series Using the EWMATCD Algorithm

To assess the feasibility of disturbance detection in sparse Landsat time series using the proposed EWMATCD algorithm, a typically sparse time series of a forest patch, wherein the average number of clear observations per year is close to 5, was used to detect disturbance (see Fig. 4). The typical change detection algorithm using the harmonic model, EWMACD, was operated to contrast with the EWMATCD algorithm. The target forest pixel for the EWMACD algorithm is located in the third row and second column of the 3 × 3 spatial window (yellow box) in Landsat images (see Fig. 4), and the first disturbance for this pixel was observed on April 22, 2009. A reliable fitting model should be established as the baseline for disturbance detection. However, fitting a harmonic model becomes a challenge when not enough temporal observations are available. When only six temporal observations were used to establish the harmonic model with two harmonic terms, the fitting values exceeded 1, and overfitting was observed even though the determination coefficient R^2 reached up to 0.99 [see Fig. 4(a)]. Overfitting influences the removal of seasonality, and the disturbance is disguised by seasonal variations. The standard deviation estimator of residuals for establishing the CLs in the EWMA chart is also inaccurate and may eventually lead to false alarms, omission errors, or detection delays. Meanwhile, the EWMA chart can accurately identify disturbances when the number of temporal observations reaches 15, which is thrice larger than the number (5) of fitting parameters in the harmonic model [see Fig. 4(b)]. However, the long time span (nearly four years, from 2001 to 2004) of the training period with 15 clear observations introduces difficulties in identifying multiple disturbances online in regions with frequent cloud covers and shadows or with frequent disturbances. When the spatiotemporal observations within the 3 × 3 spatial window at only six time steps were used to fit the SEM model, the EWMATCD algorithm, the combination of SNDVI, SEM model, and EWMA-t chart, can timely respond to the disturbance event [see Fig. 4(c)]. Therefore, the EWMATCD algorithm is suitable for detecting disturbances in regions with a low temporal frequency of clear observations.

To further illustrate the feasibility of disturbance detection using the EWMATCD algorithm, those spatiotemporal observations with different numbers of time steps (ranging from 3 to 23), which can represent typically sparse and dense time series, were used to fit the SEM model in the training period (see Fig. 5). As the time span of the training period increases, the regression RMSEs in both the training and monitoring periods before the disturbance demonstrated a slowly increasing trend. However, all RMSEs were small, peaking at 0.04 and 0.08 in the training and monitoring periods, respectively. Therefore, the low RMSEs suggest that no overfitting occurs even when sparse time series are used to fit the SEM model, thereby guaranteeing the accuracy of subsequent disturbances detection using the EWMA-t chart. The differences between the detected disturbance timing and the actual disturbance timing as observed from the reference data were almost zero regardless of the sparsity and density of the time series. However, increasing the length of the training period resulted in a nearly linear increase in computational time. Fortunately, our results show that disturbance can still be accurately detected even if sparse observations were used to establish the SEM model with less computational demands. Therefore, the proposed EWMATCD algorithm is feasible for

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TABLE I
SPATIAL ACCURACY ASSESSMENT OF DISTURBANCE DETECTION

	Changed patches	Stable patches	Total	User's accuracy
Changed patches	206	44	250	82.4%
Stable patches	26	224	250	89.6%
Total	232	268		
Producer's accuracy	88.8%	83.6%	**Overall**	86.0%

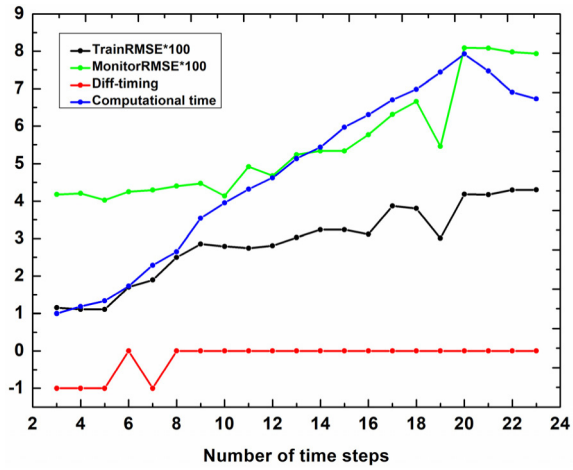


Fig. 5. Accuracy assessment of disturbance detection for the spatial patch in Fig. 4 using the EWMATCD algorithm with different numbers of time steps in the training period. TrainRMSE and MonitorRMSE represent the regression RMSE in the training and monitoring periods before a disturbance was reported. Diff-timing represents the difference between the detected disturbance timing and the actual disturbance timing as observed from the reference data. Relative computational time was defined as the ratio of computational time with a certain number of time steps in the training period to the computational time with the number of time steps in the training period equaling 3.

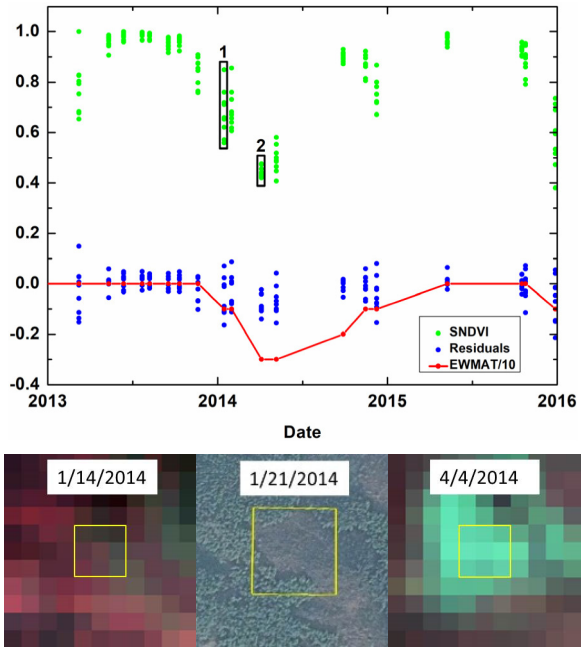


Fig. 6. Partial forest cut detection in a 3 × 3 spatial window (yellow boxes in Landsat images and high-resolution image) using the EWMATCD algorithm. The first black box in picture corresponds to the left Landsat image and middle high-resolution image, whereas the second box in picture corresponds to the right Landsat image.

disturbance detection using both sparse and dense time series, which means that it has applicability in other parts of the world with a middle or high temporal frequency of clear observations, including subtropical and tropical regions where clear observations are sparse.

B. Assessment of Disturbance Detection Accuracy in the Spatial Domain

To assess the spatial accuracy of disturbance detection, we treated both clear cuts and partial forest cuts in the forest patches with 3 × 3 pixels as disturbances. The producer's and user's accuracies for the stable patches are 83.6% and 89.6%, respectively (see Table I), whereas those for the disturbed patches are 82.4% and 88.8%. With its overall accuracy of 86.0%, the EWMATCD algorithm effectively detects disturbances.

Fig. 4(c) presents typical examples of forest clear cuts detection using the EWMATCD algorithm. The outputs of the EWMATCD algorithm and the manual interpretations also show good agreement for partial forest cuts (see Fig. 6). When the deforestation within the forest patch reaches a certain proportion in the first black box, the SNDVI values of more than half of the pixels were less than 0.7, and the partial forest cut can be detected by the EWMATCD algorithm. The EWMA-t values consistently decreased along with continued deforestation until clear cuts in the second black box, wherein the SNDVI values of all pixels were less than 0.5. The responsiveness of the EWMA-type charts to subtle changes highlights the partial forest cuts detection capability of the EWMATCD algorithm.

Most of the commission errors can be ascribed to: 1) overfitting of the SEM model and 2) the consecutive noise. When the SNDVI values for fitting the model in the training period fall within a narrow range, overfitting may still occur, thereby generating a false alarm even if the required number of time steps for SEM model estimation is lower than that for harmonic model estimation. As shown in Fig. 7(a), the data at the first five time steps for model estimation ranged between 0.9 and 1, whereas the data at the sixth time step ranged from 0.75 to 1 were considered the outliers and removed accordingly when data at these six time steps were used for model estimation. The overfitting model sent false alarms from the first black box to the third one, whose data ranged between 0.85 and 1. One solution to this problem is to properly extend the training period length, such as by increasing the number of time steps for model estimation from 6 to 9 [see Fig. 7(b)]. If noises appeared consecutively and were not captured by the Fmask algorithm, then these noises would be mistaken for

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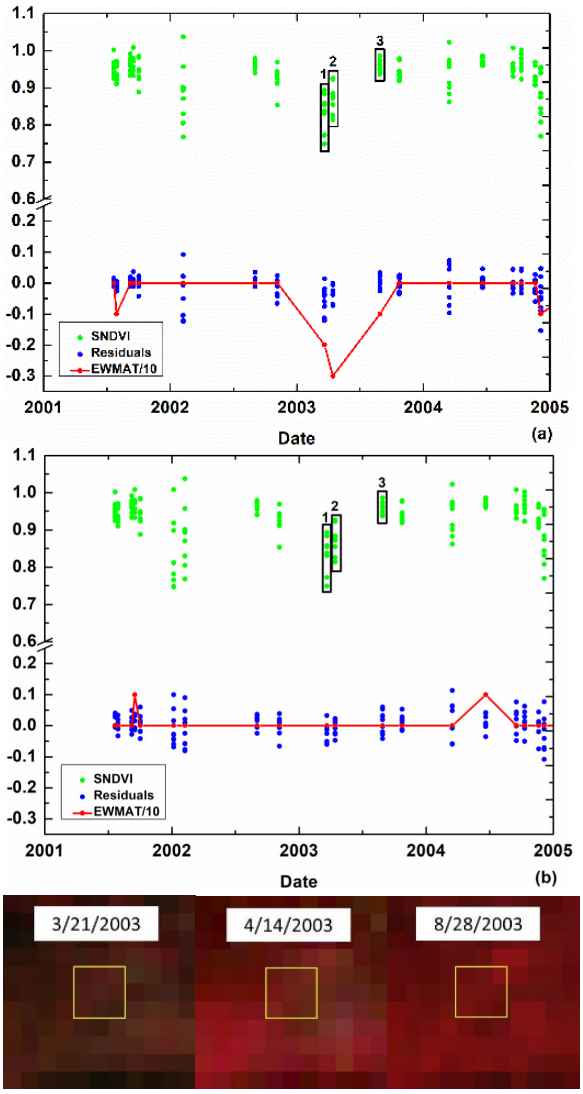


Fig. 7. Commission error in disturbance detection for the forest pixels within a 3 × 3 spatial window (yellow boxes in Landsat images) using the EWMATCD algorithm: overfitting of the SEM model. Training data in (a) six and (b) nine time steps for model estimation. The three black boxes in the picture correspond to the Landsat images.

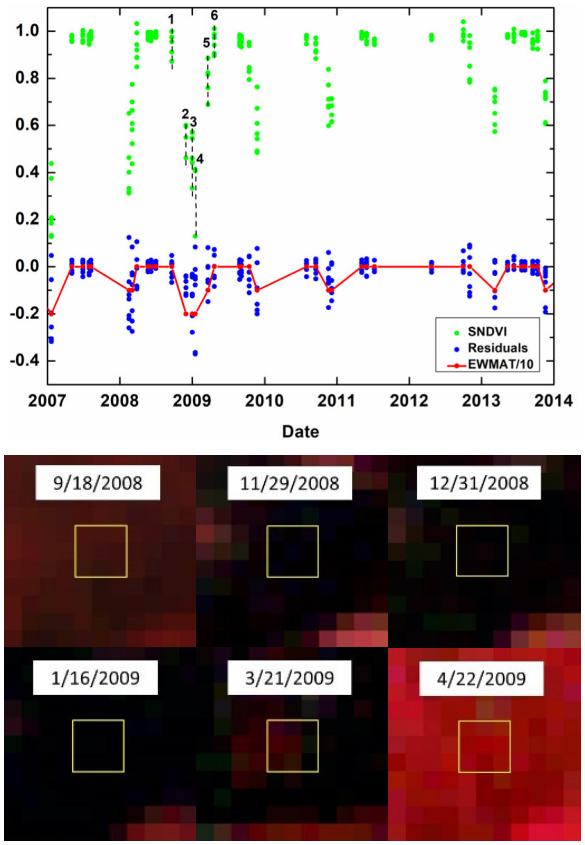


Fig. 8. Commission error in disturbance detection for the forest pixels within a 3 × 3 spatial window (yellow boxes in Landsat images) using the EWMATCD algorithm: consecutive shadows. The six dashed lines in the picture correspond to the Landsat images.

disturbance. As shown in Fig. 8, due to the mountain terrain, solar altitude, and azimuth, shadows periodically appeared in this patch at the beginning and end of each year. Although the requirement of three consecutive signaling for determining disturbance effectively removed most of the false positives due to undetected shadows, signaling still occurred over three consecutive times (from the second dashed line to the fifth one) still occurred due to the local dense distribution of time series at the end of 2008 and beginning of 2009, which was falsely detected as a disturbance.

The omission errors are mostly due to the following reasons: 1) low proportion of deforestation within the spatial window; 2) the very small window size of normalization leading to the covering up of disturbance; and 3) the low temporal frequency of clear observations in the duration of the disturbance. Although the magnitude of changes in a small part of pixels within the spatial window is significant, T_i in the EWMA-t chart, which represents the magnitude of overall change in all pixels, remains small, thereby making the partial forest cut difficult to detect. As shown in Fig. 9, clear cuts only occurred in the second pixel within the spatial window on October 16, 2015 (first black box), and deforestation did not occur in the other pixels at the subsequent time steps, such as on July 17, 2017 (second black box), thereby resulting in a small spectral change and the failure of deforestation detection with low proportion. If the spatially normalized window is smaller than the size of the disturbance event, then the focal forest pixel corresponding to $\mathrm{NDVI}_{\mathrm{pixel}}$ and the forest pixel corresponding to $\mathrm{NDVI}_{\mathrm{median}}$ are both disturbed. Therefore, the SNDVI of the focal pixel as calculated by the ratio of $\mathrm{NDVI}_{\mathrm{pixel}}$ to $\mathrm{NDVI}_{\mathrm{median}}$ is close to 1, and the influence of disturbance is smoothed out from the SNDVI time series (see Fig. 10). In this case, to minimize the possibility of selecting the NDVI value of the disturbed pixel as $\mathrm{NDVI}_{\mathrm{median}}$, appropriate normalized window size should be selected according to the range of disturbance events in the particular region. A very low temporal frequency of clear observations in the duration of disturbance may not satisfy the requirements for determining the disturbances of EWMAT values at multiple consecutive time steps less than 0. As shown in Fig. 11, only the clear observations at two time steps were observed in the longest possible duration of disturbance (January 2014 to August 2014), and two consecutive out-of-control signals were detected during this period (second and third black boxes).

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TABLE II
TEMPORAL ACCURACY ASSESSMENT OF DISTURBANCE DETECTION

	Same time step	Late = 1 time step	Late ≥ 2 time steps	Total
Changed patches	169	24	13	206
proportion	82.0%	11.7%	6.3%	100%

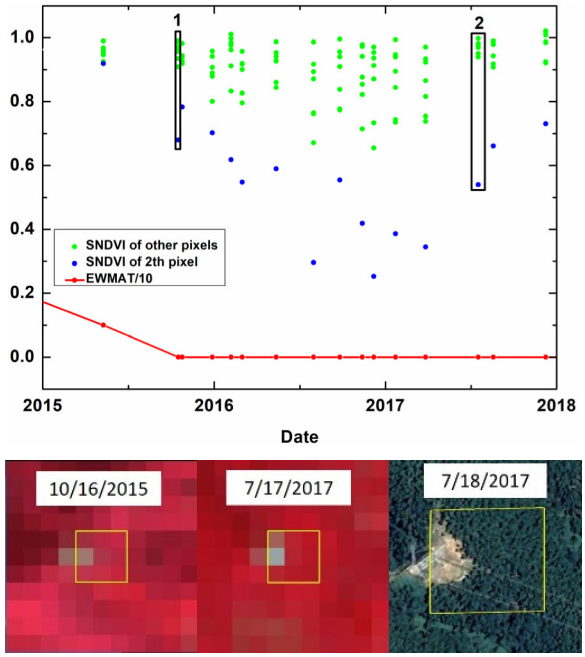


Fig. 9. Omission error in disturbance detection for the forest pixels within a 3 × 3 spatial window (yellow boxes in Landsat images and high-resolution image) using the EWMATCD algorithm: low proportion of deforestation. The two black boxes in the picture correspond to the Landsat images.

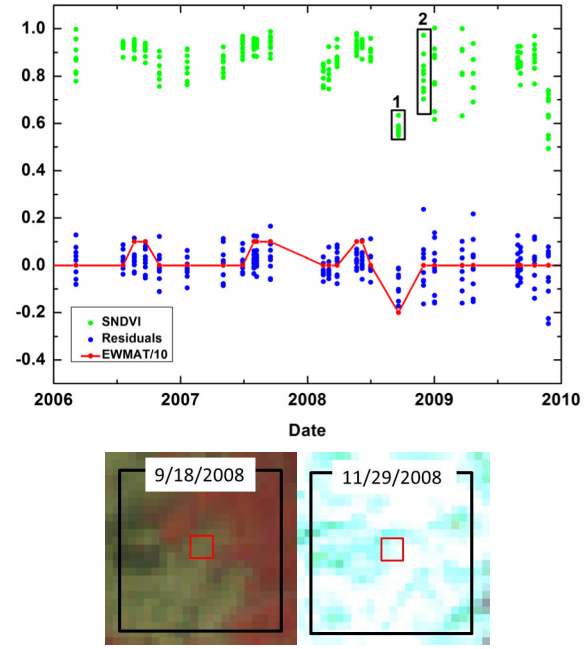


Fig. 10. Omission error in disturbance detection for the forest pixels within a 3 × 3 spatial window (red boxes in Landsat images) using the EWMATCD algorithm: very small window size of normalization leading to the covering-up of disturbance. The two black boxes in the picture correspond to the Landsat images. The black boxes in the Landsat images represent the windows for NDVI normalization (19 × 19 pixels).

However, the number of signaling remained below the required number (3) of signaling for determining disturbance according to the *persistence* index (0.5) and the mean number (5.54) of time steps per year in this patch.

C. Assessment of Disturbance Detection Accuracy in the Temporal Domain

Considering the availability of sparse Landsat time series, we treated the date corresponding to the first image regardless of whether total or partial disturbances were identified through manual visual interpretation as the occurrence date of the forest disturbance. We then further assessed the temporal accuracy of disturbance detection for 206 reference patches, wherein the disturbance was identified through both manual visual interpretation and the EWMATCD algorithm (see Table II). The proportion of patches wherein the disturbance date detected by the EWMATCD algorithm is the same as that identified using reference data was 82.0%. If we consider that detecting disturbance within two time steps of the first observable disturbance time step is acceptable, then the disturbances in 93.7% (82.0% + 11.7%) of the reference disturbance patches were correctly signaled. The EWMATCD algorithm shows a rapid response to disturbance, mostly signaling in the first two images after the disturbance.

The temporal errors are mainly due to the emergence of partial cuts with a stable proportion within the patches, which generates a small disturbing signal (T_i) at each time step. The disturbance cannot be identified by the EWMAT algorithm until the accumulated disturbing signal (EWMA-t_i) is large enough. As shown in Fig. 12, the proportion (six deforestation pixels/nine total pixels) of deforestation remained stable from the second to fourth black boxes, leading to relatively low absolute values of T_i. Therefore, the EWMA-t_i values corresponding to the second and third black boxes, which were calculated based on the low absolute values of the current T_i and the positive value of the previous T_i before deforestation (first black box), did not cross the lower CLs (−1), thereby resulting in disturbance detection delay.

D. Antinoise Capability Assessment

The antinoise capability has a key role in decreasing the false positives that reduce the influence of noise on disturbance detection. Except for the previous Fmask algorithm and the rule of consecutive signaling for determining disturbances introduced in this study, the proposed EWMATCD algorithm's high resistance to noise is mostly ascribed to the following reasons: 1) spatial normalization of NDVI time series; 2) the

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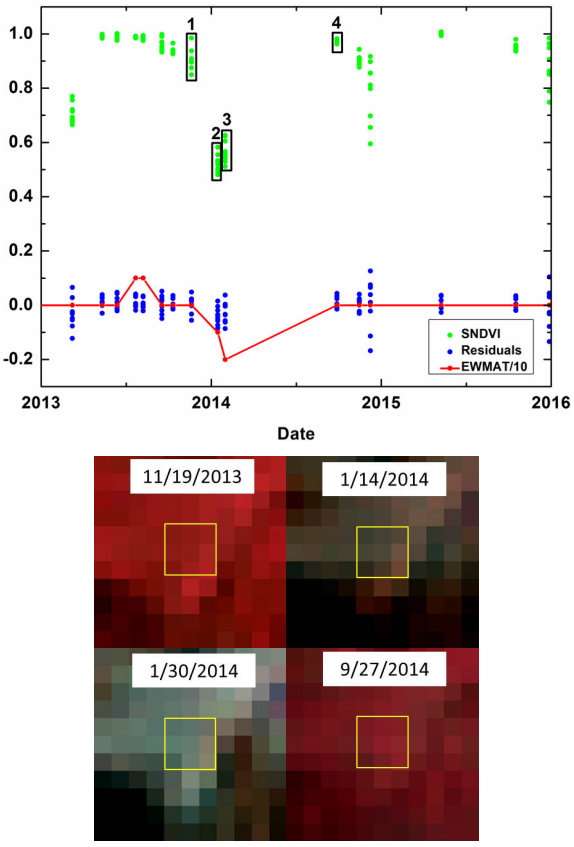


Fig. 11. Omission error in disturbance detection for the forest pixels within a 3 × 3 spatial window (yellow boxes in Landsat images) using the EWMATCD algorithm: low temporal frequency of clear observations in the duration of the disturbance. The four black boxes in the picture correspond to the Landsat images.

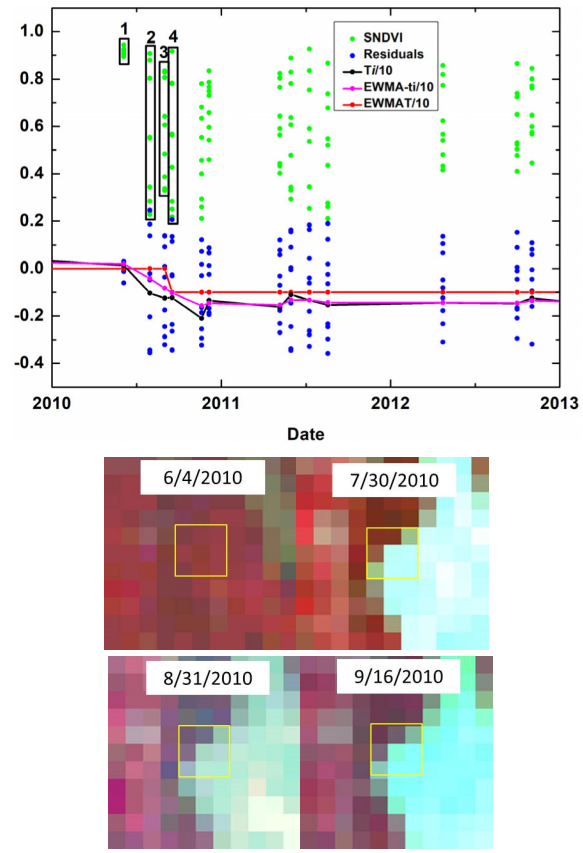


Fig. 12. Temporal error in disturbance detection for the forest pixels within a 3 × 3 spatial window (yellow boxes in Landsat images) using the EWMATCD algorithm: partial cuts with a stable proportion. The four black boxes in the picture correspond to the Landsat images.

integration of the spatial neighboring information into the temporal process (SEM model); and 3) the combined residuals of spatial neighbors by using T_i in the EWMA-t chart. Spatial normalization may reduce and even remove the effect of noise over a region that was not captured by the previous Fmask algorithm. By considering the spatial correlation within the 3 × 3 spatial window during the regression at the temporal dimension, the SEM model can reduce the influence of noise and further smoothen the regression residuals. Take the center pixel (target pixel) within the 3 × 3 spatial window in Fig. 4, for example, the mean, standard deviation, and maximum absolute values of the target pixel's residuals calculated by the SEM model [see Fig. 13(a)] are all smaller than those calculated by the harmonic regression model, which is based only on observations in the temporal domain [see Fig. 13(b)]. On the basis of smoothing the regression residuals by using spatial normalization and the SEM model, the plotting statistic T_i in the EWMA-t chart, instead of the individual regression residual for disturbance detection, was calculated by combining the residuals of spatial neighbors to further reduce the effect of potential noises or outliers within the 3 × 3 spatial window. For example, although the residuals of the target pixel on April 3, 2005 (first black box) and January 16, 2009 (fourth black box) are relatively large compared with those of most spatial neighbors, the plotting statistic T_i, the combination of the residuals of the target pixel and the small residuals of the spatial neighbors (spatial context) remains small, hence preventing the EWMA-t statistic EWMA-t_i from crossing the CLs. Similarly, the residuals of the three pixels within the spatial window on January 19, 2007 (second black box) are large, thereby allowing the EWMA-t_i to cross the CLs (EWMAT $= -1$). However, the small residuals of the other pixels (including the target pixel) at this time step (spatial context) and almost all pixels at the next several time steps (temporal context) prevent EWMA-t_i from largely deviating from the CLs and rapidly return to normal [see Fig. 13(a)], thereby preventing false alarms from being generated when the EWMA chart is executed even if the residuals of the target pixel are small at the next several time steps [see Fig. 13(b)]. In addition, up to four contaminated pixels in the spatial window were captured by the previous Fmask algorithm on February 15, 2008 (third black box). Therefore, the residuals in this box were removed to reduce the effects of potential noise [see Fig. 13(a)]. A large T_i was also avoided due to the potential noise at this time step, thereby preventing EWMA-t_i from persistently deviating from the CLs even if the residuals were normal at the following time steps, as shown in Fig. 13(b). Therefore, the proposed EWMATCD algorithm, which combines spatial and temporal contexts using spatial normalization, the SEM model, and the EWMA-t chart, is highly resistant to noises that were undetected by the Fmask algorithm.

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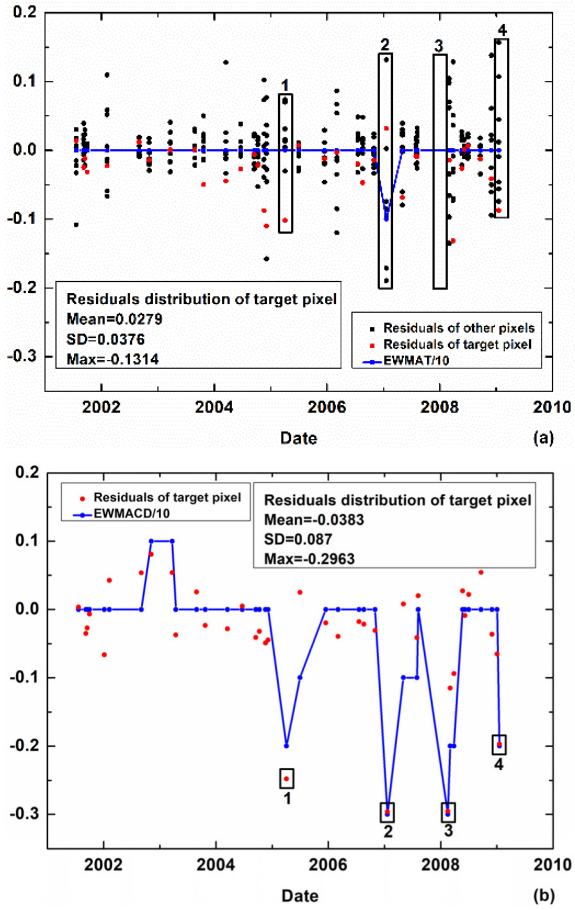


Fig. 13. Residuals of (a) SEM model for SNDVI time series within a 3 × 3 spatial window and (b) harmonic regression model for NDVI time series before disturbance.

V. Discussion

We treated all forest pixels within each 3 × 3 spatial window as a whole and defined the disturbance time at which all or part of forest pixels were disturbed. However, this treatment may result in false positives for the undisturbed pixels within the spatial window in the entire time series or reduce the temporal accuracy for the rest of the pixels that were disturbed after the defined disturbance time. Meanwhile, the disturbance map has a reduced spatial resolution with 90 m instead of 30 m. A possible solution to these problems is to record the partially disturbed forest pixels within the spatial window based on the spatial outlier detection method (such as the Shewhart X-bar chart) after detecting disturbance using the EWMATCD algorithm and then continue to detect disturbance for the rest of the undisturbed forest pixels within the spatial window until the disturbance time for each disturbed forest pixel was determined. Another alternative approach is to get a pseudolikelihood of disturbance by running the algorithm nine times for each pixel when the same pixel is artificially placed in different locations of the nine different patches. Despite improving the spatial–temporal accuracy of disturbance detection for each disturbed forest pixel and producing the disturbance map with a 30-m resolution, these approaches are computationally expensive, hence making the proposed method less attractive.

Another question is that the proposed method merely detects and records the first disturbance of forest pixels when multiple disturbances are present in the entire time series. A key element of continuous change detection is to reinitialize the fitting model for establishing a new baseline following a disturbance [7], [18], [38], [39]. Compared with the existing continuous change detection algorithms, such as CCDC or Edyn, the proposed method only needs clear observations within a 3 × 3 spatial window at a small number of time steps after each disturbance event to reinitialize the fitting model. Therefore, this algorithm has the advantage and potential to continuously detect disturbances, especially in regions with a low temporal frequency of clear observations or a short time interval among frequent disturbances. In future studies, we plan to upgrade the proposed algorithm to achieve continuous change detection.

The observations inputted into the EWMA-t chart should be sampled with fixed temporal intervals by default [31]. However, the Landsat images were collected at irregular temporal intervals because of the influence of cloud covers and shadows, which may lead to lagging change detection especially when Landsat data are sparse. Therefore, the temporal interval weight ω of smoothing parameter λ should be introduced to describe the influence of the variable sampling intervals (VSIs) on change detection based on the EWMA-t chart. For example, the temporal interval weight ω for the EWMA-t_i calculation in (11) can be set to 1 (maximum value) when the temporal interval between the previous and current observations equals the given temporal frequency of Landsat images, whereas the weight ω decreases as the temporal interval extends the given temporal frequency. The EWMA-t chart with VSI proposed by Kazemzadeh *et al.* [40] may provide a reference for estimation of the temporal interval weight ω and is, therefore, worth exploring in future studies.

The disturbance detection accuracy of the EWMA-t chart is influenced by parameters L, λ, and n, which are used to calculate the statistic EWMA-t_i and CLs CL. To accurately and timely detect disturbances, these parameters, especially λ, should be adaptively optimized for different sample sizes n in the subgroup and shift (disturbed) magnitudes [31]. For example, the values of λ should vary for the different number of forest pixels n within the 3 × 3 spatial windows or for different disturbed magnitudes, such as clear cuts and partial forest cuts. Specifically, small values of λ close to 0 increase the sensitivity of the EWMA-t chart to small process shifts, thereby increasing false positives [41] and delayed detection of large disturbances [42]. Meanwhile, increasing the value of λ close to 1 can efficiently detect large disturbances and also make the EWMA-t chart insensitive to partial forest cuts (omission errors). However, we adopted a fixed λ only designed for the situation of $n = 9$ and clear cuts for all types of n and disturbed magnitudes, thereby reducing the disturbance detection accuracy of the EWMA-t chart to some extent. An EWMA-type chart called adaptive EWMA (AEWMA) may perform well for small and large disturbances simultaneously by adaptively modifying the smoothing parameter λ [42]. Combining the AEWMA chart and t chart (i.e., AEWMA-t chart) may overcome the shortcoming that the

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EWMA-t chart with a fixed λ is only sensitive to the fixed magnitude of disturbances. This alternative is worth exploring in future research.

VI. Conclusion

In this study, we proposed an online change detection algorithm called EWMATCD that monitors forest disturbances at a subannual scale using sparse Landsat time series. Both the harmonic model and CLs are ideally established in dense time series or by using enough observations in a long training period. However, an overfitting harmonic model cannot well describe intra-annual seasonal variations, and the established CLs cannot be used as reliable baselines for disturbance detection in regions with a low temporal frequency of clear Landsat observations. To address the issues, the following three-step strategy of integrating spatial context was designed in the EWMATCD algorithm: the spatial normalized NDVI that removes seasonality, the regression residuals computed using the SEM model, and the spatial statistic t used to establish reliable CLs with a low temporal frequency of clear observations. The applicability of the proposed EWMATCD algorithm for identifying deforestation using sparse Landsat time series was clearly demonstrated. By integrating the temporal–spatial contexts, this algorithm presents a novel and useful alternative for accurately capturing the occurrence and timing of forest disturbances in regions with low data availability. This algorithm also shows advantages and potential to detect disturbances in regions with short time intervals among frequent disturbances, or in perpetually cloudy regions with short growing seasons, such as farmlands in the subtropical and tropical regions.

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References

[1] M. C. Hansen *et al.*, "High-resolution global maps of 21st-century forest cover change," *Science*, vol. 342, pp. 850–854, Nov. 2013.

[2] J. R. Townshend *et al.*, "Global characterization and monitoring of forest cover using landsat data: Opportunities and challenges," *Int. J. Digit. Earth*, vol. 5, no. 5, pp. 373–397, Aug. 2012.

[3] C. Huang, S. N. Goward, J. G. Masek, N. Thomas, Z. Zhu, and J. E. Vogelmann, "An automated approach for reconstructing recent forest disturbance history using dense landsat time series stacks," *Remote Sens. Environ.*, vol. 114, no. 1, pp. 183–198, Jan. 2010.

[4] R. E. Kennedy, Z. Yang, and W. B. Cohen, "Detecting trends in forest disturbance and recovery using yearly landsat time series: 1. LandTrendr—Temporal segmentation algorithms," *Remote Sens. Environ.*, vol. 114, no. 12, pp. 2897–2910, Dec. 2010.

[5] Z. Zhu, C. E. Woodcock, and P. Olofsson, "Continuous monitoring of forest disturbance using all available landsat imagery," *Remote Sens. Environ.*, vol. 122, pp. 75–91, Jul. 2012.

[6] E. B. Brooks, R. H. Wynne, V. A. Thomas, C. E. Blinn, and J. W. Coulston, "On-the-fly massively multitemporal change detection using statistical quality control charts and landsat data," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 6, pp. 3316–3332, Jun. 2014.

[7] E. B. Brooks, Z. Yang, V. A. Thomas, and R. H. Wynne, "Edyn: Dynamic signaling of changes to forests using exponentially weighted moving average charts," *Forests*, vol. 8, no. 9, pp. 1–18, Aug. 2017.

[8] B. DeVries, J. Verbesselt, L. Kooistra, and M. Herold, "Robust monitoring of small-scale forest disturbances in a tropical montane forest using landsat time series," *Remote Sens. Environ.*, vol. 161, pp. 107–121, May 2015.

[9] B. DeVries, M. Decuyper, J. Verbesselt, A. Zeileis, M. Herold, and S. Joseph, "Tracking disturbance-regrowth dynamics in tropical forests using structural change detection and landsat time series," *Remote Sens. Environ.*, vol. 169, pp. 320–334, Nov. 2015.

[10] L. P. Dutrieux, J. Verbesselt, L. Kooistra, and M. Herold, "Monitoring forest cover loss using multiple data streams, a case study of a tropical dry forest in Bolivia," *ISPRS J. Photogramm. Remote Sens.*, vol. 107, pp. 112–125, Sep. 2015.

[11] T. Hermosilla, M. A. Wulder, J. C. White, N. C. Coops, and G. W. Hobart, "Regional detection, characterization, and attribution of annual forest change from 1984 to 2012 using Landsat-derived time-series metrics," *Remote Sens. Environ.*, vol. 170, pp. 121–132, Dec. 2015.

[12] E. Hamunyela, J. Verbesselt, and M. Herold, "Using spatial context to improve early detection of deforestation from landsat time series," *Remote Sens. Environ.*, vol. 172, pp. 126–138, Jan. 2016.

[13] E. Hamunyela, J. Reiche, J. Verbesselt, and M. Herold, "Using space-time features to improve detection of forest disturbances from Landsat time series," *Remote Sens.*, vol. 9, no. 6, pp. 1–17, May 2017.

[14] W. B. Cohen, Z. Yang, and R. Kennedy, "Detecting trends in forest disturbance and recovery using yearly landsat time series: 2. TimeSync—Tools for calibration and validation," *Remote Sens. Environ.*, vol. 114, no. 12, pp. 2911–2924, Dec. 2010.

[15] W. B. Cohen *et al.*, "How similar are forest disturbance maps derived from different Landsat time series algorithms?" *Forests*, vol. 8, no. 4, pp. 1–19, Mar. 2017.

[16] J. Verbesselt, R. Hyndman, G. Newnham, and D. Culvenor, "Detecting trend and seasonal changes in satellite image time series," *Remote Sens. Environ.*, vol. 114, no. 1, pp. 106–115, Jan. 2010.

[17] J. Verbesselt, A. Zeileis, and M. Herold, "Near real-time disturbance detection using satellite image time series," *Remote Sens. Environ.*, vol. 123, pp. 98–108, Aug. 2012.

[18] Z. Zhu and C. E. Woodcock, "Continuous change detection and classification of land cover using all available landsat data," *Remote Sens. Environ.*, vol. 144, pp. 152–171, Mar. 2014.

[19] J. Chen, J. Chen, H. Liu, and S. Peng, "Detection of cropland change using multi-harmonic based phenological trajectory similarity," *Remote Sens.*, vol. 10, no. 7, pp. 1–20, Jun. 2018.

[20] S. Jin and S. A. Sader, "MODIS time-series imagery for forest disturbance detection and quantification of patch size effects," *Remote Sens. Environ.*, vol. 99, no. 4, pp. 462–470, Dec. 2005.

[21] D. M. Tang, H. Fan, and Y. Zhang, "Review on Landsat time series change detection methods," *J. Geo Inf. Sci.*, vol. 19, no. 8, pp. 1069–1079, Aug. 2017.

[22] E. B. Brooks, V. A. Thomas, R. H. Wynne, and J. W. Coulston, "Fitting the multitemporal curve: A Fourier series approach to the missing data problem in remote sensing analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 9, pp. 3340–3353, Sep. 2012.

[23] R. V. Platt, M. V. Ogra, R. Badola, and S. A. Hussain, "Conservation-induced resettlement as a driver of land cover change in India: An object-based trend analysis," *Appl. Geogr.*, vol. 69, pp. 75–86, Apr. 2016.

[24] L. Wu *et al.*, "Multi-type forest change detection using BFAST and monthly landsat time series for monitoring spatiotemporal dynamics of forests in subtropical wetland," *Remote Sens.*, vol. 12, no. 2, pp. 1–33, Jan. 2020.

[25] F. Gao *et al.*, "Fusing Landsat and MODIS data for vegetation monitoring," *IEEE Geosci. Remote Sens. Mag.*, vol. 3, no. 3, pp. 47–60, Sep. 2015.

[26] S. Healey, W. Cohen, Y. Zhiqiang, and O. Krankina, "Comparison of tasseled cap-based landsat data structures for use in forest disturbance detection," *Remote Sens. Environ.*, vol. 97, no. 3, pp. 301–310, Aug. 2005.

[27] T. Hilker *et al.*, "A new data fusion model for high spatial- and temporal-resolution mapping of forest disturbance based on Landsat and MODIS," *Remote Sens. Environ.*, vol. 113, no. 8, pp. 1613–1627, Aug. 2009.

[28] X. Chen *et al.*, "Cross-sensor comparisons between landsat 5 TM and IRS-P6 AWiFS and disturbance detection using integrated landsat and AWiFS time-series images," *Int. J. Remote Sens.*, vol. 34, no. 7, pp. 2432–2453, Dec. 2012.

[29] J. Reiche, E. Hamunyela, J. Verbesselt, D. Hoekman, and M. Herold, "Improving near-real time deforestation monitoring in tropical dry forests by combining dense Sentinel-1 time series with landsat and ALOS-2 PALSAR-2," *Remote Sens. Environ.*, vol. 204, pp. 147–161, Jan. 2018.

[30] C. P. Quesenberry, "SPC Q charts for start-up processes and short or long runs," *J. Quality Technol.*, vol. 23, no. 3, pp. 213–224, Feb. 2018.

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[31] L. Zhang, G. Chen, and P. Castagliola, "On t and EWMA t charts for monitoring changes in the process mean," *Qual. Rel. Eng. Int.*, vol. 25, no. 8, pp. 933–945, Dec. 2009.
[32] E. Del Castillo and D. C. Montgomery, "Short-run statistical process control: Q−chart enhancements and alternative methods," *Qual. Reliab. Eng. Int.*, vol. 10, no. 2, pp. 87–97, Mar. 2010.
[33] G. Celano, P. Castagliola, E. Trovato, and S. Fichera, "Shewhart and EWMA t control charts for short production runs," *Qual. Rel. Eng. Int.*, vol. 27, no. 3, pp. 313–326, Apr. 2011.
[34] M. A. Wulder *et al.*, "The global Landsat archive: Status, consolidation, and direction," *Remote Sens. Environ.*, vol. 185, pp. 271–283, Nov. 2016.
[35] J. Chen, X. Zhu, J. E. Vogelmann, F. Gao, and S. Jin, "A simple and effective method for filling gaps in Landsat ETM+ SLC-off images," *Remote Sens. Environ.*, vol. 115, no. 4, pp. 1053–1064, Apr. 2011.
[36] S. Jin, L. Yang, P. Danielson, C. Homer, J. Fry, and G. Xian, "A comprehensive change detection method for updating the national land cover database to circa 2011," *Remote Sens. Environ.*, vol. 132, pp. 159–175, May 2013.
[37] M. Lu, E. Pebesma, A. Sanchez, and J. Verbesselt, "Spatio-temporal change detection from multidimensional arrays: Detecting deforestation from MODIS time series," *ISPRS J. Photogram. Remote Sens.*, vol. 117, pp. 227–236, Jul. 2016.
[38] Y. Yuan *et al.*, "Continuous change detection and classification using hidden Markov model: A case study for monitoring urban encroachment onto farmland in Beijing," *Remote Sens.*, vol. 7, no. 11, pp. 15318–15339, Nov. 2015.
[39] Z. Wang *et al.*, "Continuous change detection of forest/grassland and cropland in the loess plateau of China using all available Landsat data," *Remote Sens.*, vol. 10, no. 11, p. 1775, Nov. 2018.
[40] R. B. Kazemzadeh, M. Karbasian, and M. A. Babakhani, "An EWMA t chart with variable sampling intervals for monitoring the process mean," *Int. J. Adv. Manuf. Technol.*, vol. 66, nos. 1–4, pp. 125–139, Apr. 2013.
[41] R. Saxena *et al.*, "Towards a polyalgorithm for land use change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 144, pp. 217–234, Oct. 2018.
[42] G. Capizzi and G. Masarotto, "An adaptive exponentially weighted moving average control chart," *Technometrics*, vol. 45, no. 3, pp. 199–207, Aug. 2003.



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